

CHEMISTRY

1. The empirical formula of an organic compound is CH_2O . If its exact molecular weight is determined to be 180 amu, what is its molecular formula?

- A) $\text{C}_2\text{H}_4\text{O}_2$ B) $\text{C}_3\text{H}_6\text{O}_3$ C) $\text{C}_6\text{H}_{12}\text{O}_6$ D) $\text{C}_{12}\text{H}_{22}\text{O}_{11}$

2. If 3.011×10^{23} molecules of an unknown diatomic gas have a mass of exactly 14.0 grams at standard conditions, which of the following is the gas?

- A) O_2 B) N_2 C) H_2 D) Cl_2

3. If $n=2$ and $d=1$ then total number of electron in all orbital is:

- A) 2 B) 4 C) 6 D) 10

4. Which of the following represents the correct decreasing order of ionic radii for the given isoelectronic species?

- A) $\text{Mg}^{2+} > \text{Na}^+ > \text{F}^- > \text{O}^{2-} > \text{N}^{3-}$ B) $\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+}$
 C) $\text{F}^- > \text{O}^{2-} > \text{N}^{3-} > \text{Na}^+ > \text{Mg}^{2+}$ D) $\text{Na}^+ > \text{Mg}^{2+} > \text{N}^{3-} > \text{O}^{2-} > \text{F}^-$

5. Max no. of orbitals in in 4th orbit is

- A) 6 B) 9 C) 16 D) 20

6. What is the correct oxidation number of Chromium (Cr) in chromium pentoxide (CrO_5)?

- A) +10 B) +6 C) +5 D) +4

7. According to Boyle's Law, if the volume of an ideal gas is doubled while keeping the temperature and the number of moles constant, what happens to its pressure?

- A) It doubles. B) It halves. C) It quadruples. D) It remains unchanged.

8. Body-centered cubic lattice has C.N of

- A) 4 B) 8 C) 6 D) 12

9. For the gaseous reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$, what is the proper unit of the equilibrium constant K_p ?

- A) atm B) atm^{-1} C) atm^2 D) It is unitless.

10. If 25 mL of a 0.2 N HCl solution is mixed completely with 50 mL of a 0.1 N NaOH solution, the resulting mixture will be:

- A) Strongly acidic B) Mildly acidic C) Mildly basic D) Exactly neutral

11. What is the pH of a 1.0×10^{-3} M solution of $\text{Ba}(\text{OH})_2$ at 25°C , assuming complete dissociation?

- A) 11.0 B) 11.3 C) 12.0 D) 2.7

12. The solubility product constant (K_{sp}) of silver chromate (Ag_2CrO_4) is 4.0×10^{-12} . What is its molar solubility in pure water?

- A) 1.0×10^{-4} M B) 2.0×10^{-4} M C) 1.0×10^{-6} M D) 4.0×10^{-12} M

13. Which of the following expressions correctly relates the half-life ($t_{1/2}$) of a first-order reaction to its rate constant (k)?

- A) $t_{1/2} = [\text{A}]^0 / 2k$ B) $t_{1/2} = 0.693 / k$ C) $t_{1/2} = 1 / (k \cdot [\text{A}]^0)$ D) $t_{1/2} = k / 0.693$

14. If the temperature coefficient of a specific chemical reaction is exactly 2, by how many times will the rate increase if temperature is raised from 20°C to 60°C ?

- A) 4 times B) 8 times C) 16 times D) 32 times

15. How much electrical charge is strictly required for the complete electrochemical reduction of 1 mole of MnO_4^{4-} to Mn^{2+} ?

- A) 1 Faraday B) 2 Faradays C) 5 Faradays D) 7 Faradays

16. For a given chemical reaction, $\Delta H = 30 \text{ kJ/mol}$ and $\Delta S = 100 \text{ J/(K}\cdot\text{mol)}$. At what temperature does the reaction spontaneously begin to proceed in the forward direction?

- A) Above 300 K B) Below 300 K C) Above 0.3 K D) Below 273 K

17. After exactly 3 half-lives have elapsed, what fraction of the initial amount of a radioactive isotope has decayed?

- A) 1/8 B) 1/3 C) 7/8 D) 1/4

18. Minamata disease is a severe neurological syndrome caused by the chronic toxicity of which heavy metal listed in the bio-inorganic syllabus?

- A) Arsenic (As) B) Lead (Pb) C) Mercury (Hg) D) Cadmium (Cd)

19. Which allotrope of carbon acts as a good conductor of electricity due to the presence of free, delocalized pi electrons in its layered structure?

- A) Diamond B) Fullerene C) Graphite D) Carbon monoxide

20. During the metallurgy of ores, the process of calcination is typically employed for which type of ores, and under what specific atmospheric condition?

- A) Sulfide ores in the presence of excess air B) Carbonate and hydrated oxide ores in the absence or limited supply of air
 C) Oxide ores in the presence of a carbon reducing agent. D) Halide ores under electrolytic reduction.

21. Concentrated sulfuric acid acts as a powerful dehydrating agent. When it reacts with ordinary cane sugar (sucrose), it leaves behind a porous, black mass of carbon. What is this specific chemical process called?

- A) Roasting B) Bessemerization C) Charring D) Poling

22. What is the correct chemical formula for 'white vitriol', an important chemical property of zinc mentioned in the syllabus?

- A) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ B) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ C) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ D) $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$

23. In the context of the sodium-glucose cotransporter (SGLT) mentioned in the bio-inorganic syllabus, how does this biological pump facilitate the transport of glucose into a cell against its concentration gradient?

- A) By directly hydrolyzing ATP to actively push glucose molecules into the cell.
B) By exchanging intracellular potassium ions for extracellular glucose molecules.
C) By coupling the inward 'downhill' transport of Na^+ with the inward 'uphill' transport of glucose.
D) By utilizing the calcium gradient created by the macro-nutrients.

24. According to Crystal Field Theory (CFT), when a strong field ligand like cyanide (CN^-) approaches a central transition metal ion to form an octahedral complex, what is the resulting electronic arrangement?

- A) The crystal field splitting energy (A_o) is less than the pairing energy (P), resulting in a high-spin complex.
B) The crystal field splitting energy (A_o) is greater than the pairing energy (P), resulting in forced electron pairing in the t_{2g} orbitals (low-spin complex).
C) Electrons occupy the e_g orbitals before pairing in the lower energy t_{2g} orbitals.
D) The ligand prevents d-d transitions, resulting in a colorless complex.

25. In the manufacture of steel using the Basic Oxygen Method (BOM), what is the primary metallurgical purpose of blowing high-purity oxygen at supersonic speeds over molten pig iron?

- A) To reduce the melting point of the iron by forming a molten slag blanket.
B) To act as a basic flux to neutralize the acidic silica gangue.
C) To rapidly oxidize and remove impurities like carbon, silicon, and phosphorus as volatile oxides or removable slag.
D) To dissolve into the iron lattice, transforming it into high-strength carbon steel.

26. Calomel and corrosive sublimate are two pharmacologically and chemically distinct compounds of mercury. What are their respective chemical formulas and the oxidation state of mercury in each?

- A) Calomel: HgCl_2 (+2), Corrosive Sublimate: Hg_2Cl_2 (+1) B) Calomel: Hg_2Cl_2 (+1), Corrosive Sublimate: HgCl_2 (+2)
C) Calomel: Hg_2Cl_2 (+1), Corrosive Sublimate: HgCl_2 (+2) D) Calomel: Hg_2Cl_2 (+2), Corrosive Sublimate: HgCl_2 (+1)

27. Tailing of mercury is a characteristic chemical test used to confirm the presence of ozone gas. Which compound is formed on the surface of the mercury that causes it to lose its meniscus and stick to the glass?

- A) Mercuric oxide (HgO) B) Mercurous oxide (Hg_2O)
C) Mercuric peroxide (HgO_2) D) Mercury(II) hydroxide ($\text{Hg}(\text{OH})_2$)

28. Which of the following represents the functional group of a primary amine?

- A) $-\text{NH}-$ B) $-\text{NH}_2$ C) $-\text{N}=\text{}$ D) $-\text{CONH}_2$

29. What is the general molecular formula for an open-chain, saturated alkane?

- A) C_nH_{2n} B) $\text{C}_n\text{H}_{2n-2}$ C) $\text{C}_n\text{H}_{2n+2}$ D) $\text{C}_n\text{H}_{2n}\text{O}$

30. In the industrial classification of ethanol, what does the term 'absolute alcohol' refer to?

- A) A mixture of 95.6% ethanol and 4.4% water B) Ethanol mixed with poisonous methanol
C) 100% pure, anhydrous ethanol D) Ethanol derived exclusively from the fermentation of sugar

31. What is the common name for the haloalkane with the IUPAC name trichloromethane?

- A) Iodoform B) Bromoform C) Carbon tetrachloride D) Chloroform

32. The nitration of benzene to prepare nitrobenzene requires a standard 'nitrating mixture'. What is the composition of this mixture?

- A) Concentrated HNO_3 and dilute HCl B) Dilute HNO_3 and concentrated H_2SO_4

- C) Concentrated HNO_3 and concentrated H_2SO_4 D) Concentrated HNO_3 alone

33. Which specific reagent is used in the Etard reaction for the selective partial oxidation of toluene to benzaldehyde?

- A) KMnO_4 in acidic medium B) Chromyl chloride (CrO_2Cl_2) in CS_2
C) Ozone (O_3) followed by $\text{Zn}/\text{H}_2\text{O}$ D) Carbon monoxide (CO) and HCl with AlCl_3

34. Diethyl ether is typically stored in dark, tightly sealed brown bottles to prevent it from reacting with oxygen and light to form:

- A) Highly explosive peroxides B) Poisonous phosgene gas
C) Ethanol and ethanoic acid D) Epoxides

35. Among the following carboxylic acid derivatives, which one reacts most vigorously and readily with ammonia to form an amide?

- A) Acetic anhydride B) Ethyl acetate C) Acetyl chloride D) Acetamide

36. Which of the following substituent groups exerts the strongest electron-withdrawing inductive effect (-I effect)?

- A) $-\text{OH}$ B) $-\text{F}$ C) $-\text{NO}_2$ D) NH_2

37. When propyne gas is passed through a red-hot iron tube at 873 K, it undergoes cyclic polymerization. What is the major aromatic product?

- A) Benzene B) Toluene C) 1,3,5-Trimethylbenzene (Mesitylene) D) Xylene

38. The acid-catalyzed dehydration of 1-butanol with concentrated H_2SO_4 at 170°C primarily yields which major product?

- A) 1-Butene B) 2-Butene C) Butyl hydrogen sulfate D) Dibutyl ether

39. Direct nitration of aniline using conc. HNO_3 and conc. H_2SO_4 unexpectedly yields a significant amount (47%) of m-nitroaniline. What causes this?

- A) The $-\text{NH}_2$ group is a meta-directing deactivator.
B) Steric hindrance blocks the ortho and para positions.
C) The highly acidic mixture protonates the $-\text{NH}_2$ group to form the meta-directing anilinium ion.
D) Nitration of amines proceeds via a free-radical mechanism.

40. A student attempts to prepare a Grignard reagent (RMgX) using ethanol as the solvent. The synthesis fails because:

- A) Ethanol is non-polar and cannot solvate the magnesium.
B) The acidic proton of ethanol reacts with the Grignard reagent to form an alkane.
C) Ethanol complexes with magnesium too tightly, deactivating it.
D) Ethanol oxidizes the alkyl halide.

41. When chloroform is left exposed to air and sunlight, it slowly oxidizes to produce a highly poisonous gas. The chemical name of this gas is:

- A) Chloropicrin B) Mustard gas C) Carbonyl chloride (Phosgene) D) Dichloromethane

42. An unknown alkene undergoes ozonolysis followed by reductive workup ($\text{Zn}/\text{H}_2\text{O}$) to yield an equimolar mixture of acetone (propan-2-one) and formaldehyde (methanal). What is the IUPAC name of the original alkene?

- A) 2-Methylpropene B) 2-Methyl-2-butene C) 1-Butene D) 2-Butene

43. The reaction of acetaldehyde with hydroxylamine (NH_2OH) in a weakly acidic medium yields acetaldoxime. What is the fundamental mechanism of this reaction?

- A) Electrophilic addition B) Nucleophilic substitution ($\text{S}_\text{N}2$)
C) Nucleophilic addition followed by elimination of water D) Free radical substitution

44. Maleic acid and fumaric acid are geometrical isomers of butenedioic acid. When subjected to moderate heating, what is the observed outcome?

- A) Both acids form the exact same cyclic anhydride at the same rate.
B) Fumaric acid forms an anhydride, while maleic acid sublimates.
C) Maleic acid readily forms a cyclic anhydride, whereas fumaric acid does not.
D) Neither acid can form an anhydride due to steric strain.

45. Which of the following cannot be manufactured using the solvay process?

- A) Na_2CO_3 B) NaHCO_3 C) K_2CO_3 D) CaCl_2

46. The hypnotic drug 'chloretone' is synthesized by reacting chloroform with:

- A) Propanol B) Propanone C) Methanal D) Ethanol

47. Which radioisotope is used medically to detect restricted blood circulation?

- A) Iodine-131 B) Phosphorus-32 C) Cobalt D) Sodium-24

48. Which compound fails to give a positive Lassaigne's test for nitrogen?

- A) Urea B) Hydrazine C) Aniline D) Pyridine

49. The separation technique 'atmolysis' is specifically utilized to separate:

- A) Azeotropes B) Isotopes C) Enantiomers D) Isobars

50. The indicator required in a standard permanganometric titration is:

- A) Starch B) Methyl orange C) EBT D) None

PHYSICS

51. Which of the following physical quantities is a scalar?

- A) Torque B) Impulse C) Work D) Momentum

52. The ratio of lateral strain to longitudinal strain within the elastic limit is defined as:

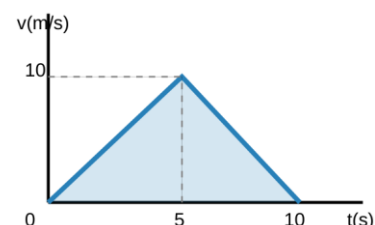
- A) Young's modulus B) Bulk modulus C) Poisson's ratio D) Shear modulus

53. The aerodynamic lift generated by an airplane's wings is primarily an application of:

- A) Archimedes' principle B) Bernoulli's principle C) Stokes' law D) Poiseuille's equation

54. The velocity-time (v-t) graph of a particle moving in a straight line is shown below. What is the total displacement of the particle in 10 seconds?

- A) 25 m B) 50 m
C) 100 m D) 0 m



55. A particle moves in a horizontal circle of radius 'r' with uniform speed 'v'.

What is the work done by the centripetal force during one complete revolution?

- A) mv^2 / r B) $(mv^2/r) \times 2\pi r$ C) Zero D) mv^2

56. If the distance between the Earth and the Moon is magically doubled, the gravitational force of attraction between them would:

- A) Double B) Halve C) Become one-fourth D) Remain the same

57. Two blocks of masses $m_1 = 2$ kg and $m_2 = 3$ kg are in contact on a frictionless horizontal table. A horizontal force $F = 10$ N is applied to m_1 as shown. What is the magnitude of the contact force between the two blocks?

- A) 10 N B) 6 N
C) 4 N D) Zero



58. A solid cylinder and a hollow cylinder of the exact same mass and same external radius are released simultaneously from the top of a rough inclined plane. Which one reaches the bottom first without slipping?

- A) Solid cylinder B) Hollow cylinder
C) Both reach simultaneously D) Depends on the angle of inclination

59. A solid piece of ice is floating in a glass of pure water. When the ice melts completely without any temperature change, the level of water in the glass will:

- A) Rise B) Fall C) Remain the same D) Overflow instantly

60. A simple pendulum has a time period T on the surface of the Earth. If the pendulum is taken to a height equal to the radius of the Earth (R) above the surface, what will be its new time period?

- A) T B) $T/2$ C) 2T D) 4T

61. Based on the working principles of different thermometers, which type is primarily used to measure extremely high temperatures (e.g., of a furnace) without requiring physical contact?

- A) Liquid-in-glass thermometer B) Resistance thermometer
C) Radiation thermometer (Pyrometer) D) Constant volume gas thermometer

62. In an isochoric thermodynamic process, which of the following variables must remain constant?

- A) Pressure B) Temperature C) Internal Energy D) Volume

63. If the absolute temperature of an ideal gas is increased by a factor of 4, the root mean square (rms) speed of its molecules will:

- A) Remain unchanged B) Increase by a factor of 2 C) Increase by a factor of 4 D) Increase by a factor of 16

64. How much quantity of heat is required to completely melt 10 grams of ice at 0°C into water at 0°C ? (Given: Latent heat of fusion of ice = 80 cal/g)

- A) 8 cal B) 80 cal C) 800 cal D) 8000 cal

65. A solid metal sphere contains a perfectly spherical hollow cavity at its center. If the metal sphere is uniformly heated to a higher temperature, the volume of the internal cavity will:

- A) Decrease, because the metal expands inward
B) Increase, because thermal expansion acts like a photographic enlargement
C) Remain completely unchanged
D) Expand or contract depending on the specific heat capacity of the metal

66. A heat engine operates between a hot reservoir at temperature T_H and a cold reservoir at temperature T_C . According to the Second Law of Thermodynamics, if the temperature of the cold reservoir (T_C) is lowered while T_H is kept constant, the maximum theoretical efficiency of the engine will:

- A) Decrease B) Remain unchanged C) Increase D) Become 100%

67. An ideal gas undergoes a perfectly adiabatic expansion, doing 50 Joules of work on its surroundings. What is the corresponding change in the internal energy of the gas system?

- A) +50 J B) -50 J C) 0 J D) Cannot be determined without knowing the specific heat ratio (γ)

68. Which of the following wave phenomena definitively proves the transverse nature of light waves?

- A) Interference B) Diffraction C) Polarization D) Refraction

69. The velocity of sound in an ideal gas at a constant temperature is independent of:

- A) The pressure of the gas B) The density of the gas C) The molar mass of the gas D) The humidity of the medium

70. In Young's double-slit experiment, if the distance between the two slits is halved and the distance from the slits to the screen is doubled, the fringe width will become:

- A) Half of its original value B) Double its original value C) Four times its original value D) Remain unchanged

71. A convex lens made of glass ($\mu_g = 1.5$) has a focal length of 20 cm in air. If it is completely immersed in water ($\mu_w = 4/3$), what is its new focal length?

- A) 20 cm B) 40 cm C) 60 cm D) 80 cm

72. A source of sound moves towards a stationary observer with a velocity equal to one-fifth of the velocity of sound. What is the percentage increase in the apparent frequency heard by the observer?

- A) 16.6% B) 20% C) 25% D) 0%

73. Two coherent light sources producing interference have intensities in the ratio 25:16. What is the ratio of maximum to minimum intensity ($I_{\max} : I_{\min}$) in the resulting interference pattern?

- A) 25:16 B) 81:1 C) 9:1 D) 41:9

74. A ray of light is incident at an angle of 60° on one face of a prism whose refracting angle is 30° . The emergent ray makes an angle of 30° with the incident ray. The emergent ray from the prism:

- A) is parallel to the incident ray B) grazes the face of emergence
C) is perpendicular to the face of emergence D) undergoes total internal reflection inside

75. Two polaroids are placed in the path of an unpolarized beam of intensity I_0 such that their transmission axes are crossed (90° apart). A third polaroid is introduced between them such that its axis makes an angle of 30° with the first polaroid. What is the final intensity of the transmitted light?

- A) $3 I_0 / 32$ B) $I_0 / 8$ C) $3 I_0 / 16$ D) $I_0 / 16$

76. Two point charges placed in a medium of dielectric constant $K=4$ at a specific distance experience an electrostatic force F . If the medium is completely removed (replaced by a vacuum) while keeping the charges and distance the exact same, the new force between them will be:

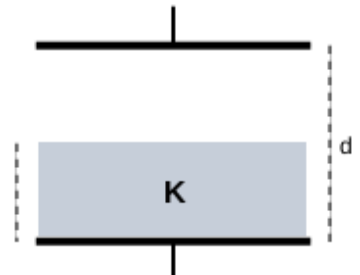
- A) $F/4$ B) F C) $4F$ D) $16F$

77. A positive test charge $+q$ is moved from point A to point B along a perfectly semicircular path of radius r . The center of this semicircle contains a stationary positive point charge $+Q$. The work done by the electrostatic force during this movement is:

- A) $\pi r \cdot (kq/r^2)$ B) kq/r C) $-kq/r$ D) Zero

78. A parallel plate capacitor has plate area A and a separation distance d between its plates. A dielectric slab of dielectric constant K and precise thickness $d/2$ is inserted between the plates, covering the full area A as shown in the figure. What is the equivalent capacitance of this system?

- A) $(K\epsilon_0 A)/d$ B) $(2K\epsilon_0 A)/(d(K+1))$
C) $(K\epsilon_0 A)/(2d)$ D) $(\epsilon_0 A(K+1))/(2d)$



79. A $4 \mu\text{F}$ capacitor is fully charged using a 50 V DC supply and then completely disconnected from the source. It is subsequently connected in parallel to an uncharged $1 \mu\text{F}$ capacitor. The total electrostatic energy lost in the form of heat and radiation during this charge redistribution is:

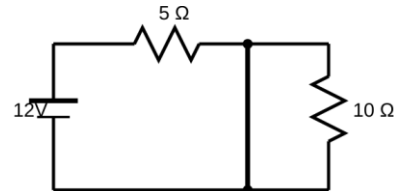
- A) $1.0 \times 10^{-3} \text{ J}$ B) $2.5 \times 10^{-3} \text{ J}$ C) $4.0 \times 10^{-3} \text{ J}$ D) $5.0 \times 10^{-4} \text{ J}$

80. A uniform cylindrical wire has a resistance R . If the wire is uniformly stretched such that its total length is exactly doubled, what will be its new resistance? (Assume the total volume of the wire remains constant).

- A) $R/2$ B) $2R$ C) $4R$ D) $8R$

81. In the electrical circuit diagram shown below, calculate the steady-state current flowing through the 10Ω resistor.

- A) 0 A B) 0.8 A
C) 1.2 A D) 2.4 A



82. When an electric current is passed through a junction of two dissimilar metals, heat is either absorbed or evolved depending on the direction of the current. This specific phenomenon is known as:

- A) Seebeck effect B) Joule heating C) Peltier effect D) Thomson effect

83. A coil has an internal resistance of 10Ω and an inductance of $0.1/\pi \text{ H}$. It is connected in series to an alternating current supply of 200 V , 50 Hz . Calculate the average electrical power entirely dissipated in this circuit.

- A) 1000 W B) 2000 W C) 4000 W D) 2828 W

84. A specimen material is found to have a relative magnetic permeability $\mu_r = 0.9999$. If this piece of material is placed in a strongly non-uniform magnetic field, it will inherently tend to:

- A) Rapidly move towards the strongest part of the magnetic field.
B) Slowly move from the stronger part to the weaker part of the field.
C) Align perfectly parallel to the direction of the magnetic field lines.
D) Remain completely unaffected and stationary.

85. According to the Biot-Savart Law (and Ampere's Law), the magnitude of the magnetic field B at a perpendicular distance r from an infinitely long, straight current-carrying conductor is proportional to:

- A) $1/r^2$ B) r C) $1/r$ D) r^2

86. A proton (charge +e) enters a uniform magnetic field of 2.0 T at an angle of 30° relative to the direction of the magnetic field lines, traveling with a velocity of 1.0×10^5 m/s. The magnitude of the magnetic Lorentz force experienced by the proton is:

- A) 3.2×10^{-14} N B) 1.6×10^{-14} N C) 2.77×10^{-14} N D) Zero

87. A strong bar magnet is dropped vertically downwards along its axis through the center of a horizontally held copper ring. As the magnet falls through the ring, its acceleration a (ignoring air resistance) will be:

- A) Exactly equal to the acceleration due to gravity ($a=g$).
B) Greater than gravity ($a>g$) as it approaches the ring.
C) Less than gravity ($a<g$) for the entire duration of the fall.
D) Less than gravity ($a<g$) while entering, but greater than gravity ($a>g$) while exiting.

88. The electric current flowing in an ideal inductor of 50 mH steadily increases from 0 A to 2.0 A over a period of 0.1 s. What is the total magnetic potential energy stored in the inductor at the end of this time interval?

- A) 0.1 J B) 0.05 J C) 1.0 J D) 0.2 J

89. Which of the following properties of a nucleus is approximately constant across different elements from iron to uranium?

- A) Binding energy B) Nuclear mass C) Nuclear density D) Packing fraction

90. The work function of a metal surface is 4.0 eV. If photons of energy 3.5 eV incident upon this surface, the kinetic energy of the emitted photoelectrons will be:

- A) 0.5 eV B) 3.5 eV C) 7.5 eV D) Photoelectric emission will not occur

91. According to Hubble's Law, the recessional velocity of a distant galaxy is:

- A) Directly proportional to the cube root of its distance.
B) Directly proportional to its distance from the observer.
C) Inversely proportional to the square of its distance.
D) Independent of its distance but dependent on its age.

92. A radioactive substance has a half-life of 20 days. The time required for 7/8 of the substance to disintegrate is:

- A) 40 days B) 60 days C) 80 days D) 120 days

93. In the biasing of a p-n junction diode, forward biasing means:

- A) Positive terminal of battery is connected to p-type and negative to n-type.
B) Positive terminal of battery is connected to n-type and negative to p-type.
C) Both p-type and n-type are connected to the ground.
D) Both p-type and n-type are connected to the same positive potential.

94. Which specific information is obtained from the J.J. Thomson's experiment regarding electrons?

- A) Charge of the electron (e) B) Mass of the electron (m)
C) Specific charge of the electron (e/m) D) Energy of the electron

95. The production of characteristic X-rays in an X-ray tube is due to:

- A) Bremsstrahlung of incident electrons. B) Transitions of inner-shell electrons in the target atoms.
C) Interaction of electrons with nuclear fields only. D) Acoustic waves generated upon electron impact.

96. Which combination of logic gates results in a circuit where the output is HIGH (1) if and only if both inputs are DIFFERENT?

- A) AND gate followed by an OR gate. B) NAND gate followed by an AND gate.
C) An XOR (Exclusive OR) structure built from fundamental gates. D) An OR gate followed by a NOT gate.

97. In a photoelectric effect experiment, keeping the frequency of incident radiation constant, if the intensity of the light is doubled, what happens to the maximum kinetic energy of the photoelectrons?

- A) It is doubled. B) It becomes four times. C) It is halved. D) It remains unchanged.

98. A radioactive sample starts with N_0 active nuclei. If the mean life is τ , what fraction of the sample will have disintegrated after time $t = \tau$?

- A) $1/e$ B) $1 - 1/e$ C) $1/2$ D) $1 - 1/\ln(2)$

99. An electron is accelerated through a non-relativistic potential difference V . What is the expression for $\lambda_{\text{deB}}/\lambda_{\text{photon}}$?

- A) $\sqrt{(2m/eV)}$ B) $\sqrt{(2meV)/hc}$ C) $\sqrt{(hc/2eV)}$ D) $\sqrt{(eV/2mc^2)}$

100. Given exact masses of neutron and proton, calculate the total binding energy of a Carbon-12 nucleus.

- A) 89.28 MeV B) 92.16 MeV C) 108.99 MeV D) 93.15 MeV

Zoology

101. Jean-Baptiste Lamarck proposed the first cohesive theory of evolution. According to the strict principles of Lamarckism, the long neck of the modern giraffe evolved because:

- A) A random genetic mutation caused long necks, giving those individuals a survival advantage during droughts.
B) Ancestral giraffes continuously stretched their necks to reach high leaves, and this acquired elongation was passed directly to their offspring.
C) Shorter-necked giraffes migrated to different environments, isolating the tall population.
D) God continuously created taller variants to match the height of growing trees.

102. Archaeopteryx lithographica is a famous paleontological fossil representing a crucial 'connecting link' in evolution. It conclusively demonstrates the evolutionary transition between which two vertebrate classes?

- A) Amphibia and Reptilia B) Reptilia and Aves (Birds) C) Aves and Mammalia D) Pisces and Amphibia

103. Following the evolutionary trend of human cranial capacity, which of the following prehistoric humans possessed a cranial capacity (approx. 1400 - 1600 cc) that was equal to or even slightly larger than that of modern Homo sapiens?

- A) Homo habilis B) Homo erectus
C) Cro-Magnon man (early Homo sapiens sapiens) D) Neanderthal man (Homo neanderthalensis)

104. The defining diagnostic feature of the Phylum Coelenterata (Cnidaria) is the presence of specialized stinging cells used for prey capture and defense. The actual explosive, microscopic stinging capsule within this cell is called the:

- A) Cnidoblast B) Nematocyst C) Colloblast D) Statocyst

105. Parasitic flatworms (like Taenia solium) exhibit profound morphological adaptations to survive inside the host's harsh digestive tract. Which of the following is a primary adaptation of adult tapeworms?

- A) A highly developed, complex nervous system for navigation.
B) The presence of a tough, enzyme-resistant syncytial tegument and total loss of the alimentary canal.
C) A highly efficient respiratory system utilizing gills.
D) Locomotory cilia covering their entire body.

106. While most mollusks possess an open circulatory system, which specific class of highly active, intelligent mollusks has evolved a highly efficient closed circulatory system?

- A) Gastropoda (Snails) B) Bivalvia (Clams)
C) Cephalopoda (Squids and Octopuses) D) Polyplacophora (Chitons)

107. A vertebrate has dry, cornified skin, epidermal scales, and respire strictly via lungs. However, unlike most of its class which possess a 3-chambered heart, this specific animal order boasts a complete 4-chambered heart. This animal belongs to the order:

- A) Squamata (Snakes) B) Chelonia (Turtles) C) Crocodilia (Crocodiles) D) Anura (Frogs)

108. In mammalian infants and hibernating animals, a specialized type of adipose tissue generates intense heat (non-shivering thermogenesis) to prevent hypothermia. The unique color and heat-generating capability of this tissue are due to massive numbers of mitochondria containing the uncoupling protein 'thermogenin'. This tissue is:

- A) White adipose tissue B) Yellow adipose tissue C) Brown adipose tissue D) Areolar tissue

109. The flat bones of the human skull (e.g., parietal and frontal bones) do not develop from a prior cartilage model. Instead, they develop directly from sheets of mesenchymal connective tissue. This developmental process is termed:

- A) Endochondral ossification B) Intramembranous ossification C) Osteoporosis D) Chondrogenesis

110. Which specific histological feature structurally differentiates skeletal muscle fibers from smooth muscle fibers under light microscopy?

- A) The presence of centrally located, single nuclei. B) The presence of a sarcolemma.
C) The presence of distinct, regular transverse striations (A and I bands). D) The ability to undergo hypertrophy.

111. In the microscopic structure of a myelinated motor neuron, the myelin sheath is not a continuous tube but is interrupted at regular intervals. These unmyelinated gaps, rich in voltage-gated sodium channels, are specifically termed:

- A) Synaptic clefts B) Dendritic spines C) Nodes of Ranvier D) Axon hillocks

112. In severe infections of Plasmodium falciparum, a rare but fatal complication involves massive intravascular hemolysis, resulting in the excretion of dark, hemoglobin-rich urine. This specific condition is clinically known as:

- A) Cerebral malaria B) Algid malaria C) Blackwater fever D) Quartan fever

113. In the sporogonic (sexual) cycle of Plasmodium within the female Anopheles mosquito, the motile ookinete penetrates the stomach wall and encysts on the outer surface. This resulting encysted stage, which undergoes massive meiotic and mitotic division, is the:

- A) Oocyst B) Sporozoite C) Trophozoite D) Cryptomerozoite

114. In the digestive system of the earthworm (Pheretima), the prominent absorptive structure called the typhlosole originates abruptly in which specific segment?

- A) Segment 14 B) Segment 15 C) Segment 26 D) Segment 95

115. Pheretima is monoecious (hermaphroditic). Which specific segments contain the two pairs of testes responsible for producing male gametes?

- A) Segments 8 and 9 B) Segments 10 and 11 C) Segments 14 and 15 D) Segments 17 and 19

116. In the anatomy of the frog's ear, a complete lack of an external ear (pinna) is observed. The eardrum (tympanum) is located on the surface. Internally, it is connected to the inner ear by a single rod-like bone called the:

- A) Malleus B) Incus C) Stapes D) Columella auris

117. A zoology student is dissecting the cranial nerves of a frog (Rana). How many pairs of cranial nerves will the student definitively identify emerging from the frog's brain?

- A) 8 pairs B) 10 pairs C) 12 pairs D) 14 pairs

118. Cholecystokinin (CCK) is a vital digestive hormone secreted by the duodenal mucosa in response to the entry of fatty chyme. What are its two primary physiological actions?

- A) Stimulating gastric acid secretion and closing the pyloric sphincter.
B) Stimulating bicarbonate secretion from the pancreas and inhibiting bile production.
C) Inducing vigorous contraction of the gallbladder and stimulating the pancreas to release enzyme-rich juice.
D) Stimulating intestinal peristalsis and releasing intrinsic factor.

119. The absorption of Vitamin B12 (Cobalamin) is a highly specialized process. Even if intrinsic factor is present, B12 can ONLY be absorbed by specific receptors located exclusively in the:

- A) Stomach lining B) Duodenum C) Jejunum D) Terminal ileum

120. In respiratory physiology, carbon monoxide (CO) is exceptionally toxic because it binds to hemoglobin with an affinity approximately 200-250 times greater than oxygen. Furthermore, CO binding causes the oxygen-hemoglobin dissociation curve to:

- A) Shift extremely to the right, dumping all remaining oxygen.
- B) Shift extremely to the left, preventing the release of any bound oxygen to the tissues.
- C) Become linear.
- D) Reverse its shape entirely.

121. A critical regulatory reflex prevents the extreme over-inflation of the lungs during forceful inspiration. Stretch receptors in the bronchioles fire inhibitory signals to the medullary respiratory centers to stop inspiration. This reflex is termed the:

- A) Bainbridge reflex
- B) Hering-Breuer reflex
- C) Baroreceptor reflex
- D) Chemoreceptor reflex

122. In the complete cardiac cycle, the physiological period when the ventricles are completely closed chambers (both AV and semilunar valves are closed) and the ventricular pressure drops precipitously as the muscle relaxes is called:

- A) Isovolumetric contraction
- B) Ventricular ejection
- C) Isovolumetric relaxation
- D) Rapid ventricular filling

123. In the capillary beds, fluid is forced out at the arterial end by hydrostatic pressure. However, fluid is drawn back into the capillary at the venous end primarily due to the osmotic pull of plasma proteins (principally albumin). This opposing force is specifically termed:

- A) Blood Colloid Osmotic Pressure (Oncotic pressure)
- B) Interstitial Fluid Hydrostatic Pressure
- C) Capsular Hydrostatic Pressure
- D) Active transport potential

124. In the kidney, an individual nephron contains a specialized capillary bed uniquely situated between two arterioles (the afferent and efferent). This high-pressure capillary bed, dedicated strictly to ultrafiltration, is the:

- A) Vasa recta
- B) Peritubular capillaries
- C) Glomerulus
- D) Macula densa

125. A patient's blood gas analysis shows a blood pH of 7.25 (acidosis) with an elevated $p\text{CO}_2$. To physiologically compensate for this respiratory acidosis, the kidneys will actively:

- A) Secrete more bicarbonate ions (HCO_3^-) into the urine.
- B) Secrete more hydrogen ions (H^+) into the urine and reabsorb new bicarbonate.
- C) Decrease the production of ammonia.
- D) Dilate the afferent arterioles.

126. During a surgical procedure, local anesthetics (like lidocaine) are utilized to block pain. They function at the cellular level by binding to and physically blocking which specific axonal structures?

- A) Voltage-gated potassium (K^+) channels
- B) Sodium-potassium (Na^+/K^+) ATPase pumps
- C) Voltage-gated sodium Na^+ channels
- D) Ligand-gated calcium (Ca^{2+}) channels

127. In the functional anatomy of the human brain, the Broca's area is a highly specialized region critical for the physical articulation and production of speech. In most individuals, it is located in the:

- A) Left frontal lobe
- B) Right parietal lobe
- C) Left occipital lobe
- D) Right temporal lobe

128. The retina contains a tiny, specialized yellowish area called the macula lutea. At its exact center is a depression that provides the absolute highest visual acuity (sharpness). This depression, packed exclusively with cone cells, is the:

- A) Optic disc (Blind spot)
- B) Fovea centralis
- C) Cornea
- D) Ciliary body

129. The adrenal cortex produces three classes of steroid hormones. The outermost layer, the zona glomerulosa, is exclusively responsible for synthesizing which life-sustaining hormone?

- A) Cortisol (Glucocorticoid)
- B) Aldosterone (Mineralocorticoid)
- C) Dehydroepiandrosterone (Androgen)
- D) Epinephrine (Catecholamine)

130. The hypothalamus controls the anterior pituitary via releasing and inhibiting hormones. Somatostatin, produced by the hypothalamus, exerts a powerful inhibitory effect preventing the release of:

- A) Prolactin
- B) Thyroid Stimulating Hormone (TSH)
- C) Growth Hormone (GH)
- D) Adrenocorticotrophic Hormone (ACTH)

131. In the process of human fertilization, the sperm must penetrate the thick glycoprotein coat surrounding the ovum. This coat is dissolved by hydrolytic enzymes (like hyaluronidase and acrosin) released from the sperm's acrosome. The acrosome is a modified version of which cellular organelle?

- A) Mitochondrion B) Lysosome / Golgi complex C) Smooth Endoplasmic Reticulum D) Centriole

132. During human embryonic development, the blastocyst implants into the uterine wall. The blastocyst consists of an inner cell mass (which becomes the embryo) and an outer cellular layer that will eventually form the fetal portion of the placenta. This outer layer is the:

- A) Trophoblast B) Endoderm C) Amnion D) Yolk sac

133. The pathogenesis of Tuberculosis (TB) involves the bacteria surviving and multiplying inside macrophages. In response, the immune system walls off the infection, creating a distinct nodular lesion in the lung tissue characterized by a core of 'caseous' (cheese-like) necrosis. This lesion is termed a:

- A) Gumma B) Granuloma (Tubercle) C) Chancre D) Carbuncle

134. Among the major hepatitis viruses (A, B, C, D, E) that primarily infect the liver, Hepatitis B is highly unique in terms of its genome because it is the only one that is a:

- A) Single-stranded RNA virus B) Double-stranded DNA virus C) Retrovirus D) Prion

135. In immunology, 'opsonization' is a highly critical process that drastically enhances the efficiency of the immune system. Opsonization is best defined as the:

- A) Direct lysis of a bacterial cell by the membrane attack complex.
B) Coating of a pathogen with antibodies or complement proteins to mark it for rapid phagocytosis.
C) Secretion of histamine by mast cells to cause vasodilation.
D) Clumping together of soluble antigens into an insoluble precipitate.

136. The principle of vaccination is based entirely on the property of 'immunological memory'. When an individual is given a booster shot years later, the immune system mounts a rapid, massive response characterized primarily by the production of:

- A) IgM antibodies B) IgG antibodies C) IgE antibodies D) Cytotoxic T-cells only

137. In the industrial production of dairy products, the fermentation of lactose into lactic acid causes milk proteins (casein) to coagulate, forming curd. Which specific microbe is universally employed as the starter culture for this process?

- A) *Saccharomyces cerevisiae* B) *Escherichia coli* C) *Lactobacillus acidophilus* D) *Penicillium notatum*

138. In advanced medical research, transgenic 'knockout' mice are incredibly valuable models for studying human diseases. A 'knockout' mouse is specifically engineered by:

- A) Inserting a foreign human gene into its genome to see if it is expressed.
B) Selectively disabling or disrupting a specific existing gene to observe the resulting phenotypic defect.
C) Cloning the mouse from a single somatic cell.
D) Treating the mouse with high-dose radiation to induce random mutations.

139. The Minamata disaster in Japan brought global attention to the dangers of industrial pollution and biomagnification. The severe neurological disease observed in the local population was caused by the consumption of fish heavily contaminated with:

- A) Radioactive Strontium-90 B) Dichlorodiphenyltrichloroethane (DDT) C) Methylmercury D) Lead

140. Nepal has established numerous protected areas to conserve its unique biodiversity. The pristine alpine habitat of the Shey Phoksundo National Park is considered a critical stronghold for the conservation of which highly elusive, endangered predator?

- A) The Royal Bengal Tiger (*Panthera tigris*) B) The Snow Leopard (*Panthera uncia*)
C) The One-horned Rhinoceros (*Rhinoceros unicornis*) D) The Asiatic Elephant (*Elephas maximus*)

BOTANY

141. Which of the following best describes the primary biological role of enzymes in living organisms?

- A) Providing primary structural support to plant cells B) Acting as the main storage form of energy
C) Lowering the activation energy required for biochemical reactions D) Carrying genetic information across generations

142. A lipid molecule is found to be solid at room temperature. This physical state is most likely due to:

- A) A high proportion of unsaturated fatty acids
B) The absence of double bonds between carbon atoms in its fatty acid chains
C) The presence of a phosphate group attached to the glycerol backbone
D) A complex multi-ring steroid structure

143. According to the 5-Kingdom classification system, which organism is correctly matched with its kingdom?

- A) Cyanobacteria - Protista B) Yeast - Monera C) Mucor - Fungi D) Spirogyra – Plantae

144. A farmer notices a severe nitrogen deficiency in the soil of a rice paddy. Which organism from the syllabus would be most beneficial to introduce to naturally address this?

- A) Dryopteris B) Cyanobacteria C) Mucor D) Spirogyra

145. In the floral formula of the family Fabaceae, the androecium is typically represented as:

- A) A(9)+1 B) A9+1 C) A(10) D) A 5+5

146. Which organism bears gemma cup?

- A) Nastoc B) Mucor C) Marchantia D) Pteridium

147. The medicinally important alkaloid reserpine is obtained from:

- A) Azadirachta indica B) Ocimum sanctum C) Rauwolfia serpentina D) Zingiber officinale

148. In Marchantia, the dominant and independent phase of the life cycle is:

- A) Sporophyte B) Zygote C) Gametophyte D) Spore mother cell

149. Phyllacode and cladode are morphological adaptation for photosynthesis and water storage, found in:

- A) Hydrophytes B) Mesophytes C) Xerophytes D) Halophytes

150. Double fertilization in angiosperms results in the formation of:

- A) Embryo and fruit B) Embryo and endosperm C) Seed coat and embryo D) Ovule and pollen grain

151. The concept of binomial nomenclature, the formal system of naming species, was formally introduced by:

- A) R.H. Whittaker B) Carl Woese C) Carolus Linnaeus D) Aristotle

152. The primary consumers in the limnetic zone of a pond are mainly:

- A) Phytoplankton B) Zooplankton C) Rooted hydrophytes D) Decomposers

153. During the process of hydrosere (ecological succession in a water body), which stage immediately follows the submerged plant stage?

- A) Reed-swamp stage B) Floating plant stage C) Marsh-meadow stage D) Woodland stage

154. The greenhouse effect, leading to climate change, is primarily caused by the atmosphere's ability to:

- A) Reflect high-energy ultraviolet radiation
B) Transmit long-wave infrared radiation directly to space
C) Absorb and re-radiate long-wave infrared radiation back to Earth
D) Absorb incoming short-wave visible light completely

155. In Kathmandu valley, phytochemical smog is seen due to:

- A) Ozone + PAN + NO_x B) Smoke + SO₂ C) Ozone + SO₂ D) Smoke + SO₂ + PAN

156. Lysosomes are formed from:

- A) Nucleus B) Endoplasmic reticulum C) Golgi apparatus D) Mitochondria

157. A chemical inhibitor that specifically blocks the assembly of spindle fibers (microtubules) would most directly halt the cell cycle during:

- A) S phase B) Prophase C) Metaphase D) Cytokinesis

158. Which of the following structures is present in both prokaryotic and eukaryotic cells?

- A) Mitochondria B) Golgi body C) Ribosome D) Chloroplast

159. The biological significance of meiosis includes all of the following EXCEPT:

- A) Maintenance of a constant chromosome number across successive generations
B) Production of genetic variation through crossing over
C) Repair of damaged somatic tissues and vegetative growth
D) Independent assortment of maternal and paternal chromosomes

160. The concept of Amitosis is best described as:

- A) A complex nuclear division involving spindle formation
B) A direct cell division without the appearance of chromosomes or spindle fibers
C) The division of germ cells to form haploid gametes
D) The synthesis phase of the cell cycle

161. Color blindness in humans is an example of:

- A) Y-linked inheritance B) Autosomal dominant inheritance C) X-linked recessive inheritance D) Codominance

162. In a case of incomplete dominance (e.g., flower color in *Mirabilis jalapa*), the phenotypic ratio of the F₂ generation in a monohybrid cross is:

- A) 3:1 B) 1:2:1 C) 9:3:3:1 D) 1:1

163. Turner's syndrome is a genetic disorder resulting from aneuploidy. Which of the following karyotypes represents this condition?

- A) 47, XXY B) 45, XO C) 47, +21 D) 47, XYY

164. According to the concept of the central dogma of molecular biology, genetic information flows sequentially from:

- A) RNA to DNA to Protein B) DNA to RNA to Protein C) Protein to RNA to DNA D) RNA to Protein to DNA

165. Which of the following genetic disorders is inherited as an autosomal recessive trait?

- A) Hemophilia B) Color blindness C) Albinism D) Down's syndrome

166. Polyploidy (originating from chromosome non-disjunction) can be artificially induced in plants using a chemical that inhibits spindle fiber formation. Which chemical is typically used?

- A) Colchicine B) Auxin C) Cytokinin D) Ethylene

167. In the transverse section (T.S.) of a typical dicot root, the arrangement of primary xylem is described as:

- A) Conjoint and collateral B) Radial and exarch C) Conjoint and closed D) Radial and endarch

168. Which specific type of plant tissue is primarily responsible for secondary growth, thereby increasing the girth (diameter) of a dicot stem?

- A) Apical meristem B) Intercalary meristem C) Lateral meristem (Cambium) D) Permanent parenchyma

169. Closed vascular bundles are commonly found in:

- A) Dicot stem B) Dicot root C) Monocot stem D) Gymnosperm stem

170. A plant cell with an osmotic potential (solute potential) of -10 bars and a turgor pressure of 4 bars is placed in a solution with a water potential of -8 bars. What will be the net direction of water movement?

- A) Water will move from the solution into the cell B) Water will move from the cell into the solution
C) There will be no net movement of water D) The cell will immediately undergo plasmolysis and burst

171. During the light-independent reactions (Calvin-Benson cycle) of photosynthesis, the very first stable organic compound formed is:

- A) Phosphoenolpyruvate (PEP) B) Ribulose 1,5-bisphosphate (RuBP)
C) 3-Phosphoglyceric acid (3-PGA) D) Oxaloacetic acid (OAA)

172. During the ascent of sap, the continuous, unbroken column of water within the narrow xylem vessels is maintained against gravity primarily by:

- A) Active transport of ions by root hairs B) Capillary action alone
C) The cohesive and adhesive properties of water molecules D) Continuous pumping by living parenchyma cells

173. Which plant growth promoter is primarily responsible for inducing apical dominance, suppressing the growth of lateral buds?

- A) Gibberellin B) Cytokinin C) Auxin D) Absciscic acid

174. In plants adapted to hot, dry environments (C4 plants), the initial fixation of carbon dioxide occurs in the:

- A) Mesophyll cells B) Bundle sheath cells C) Epidermal cells D) Guard cells

175. The physiological process by which intact plants exude water in the form of liquid droplets from hydathodes at the margins of leaves is known as:

- A) Transpiration B) Imbibition C) Guttation D) Osmosis

176. In angiosperms, the endosperm tissue is typically triploid (3n). This unique ploidy level is a direct result of the fusion between:

- A) Two male gametes and one egg cell B) One male gamete and the diploid secondary nucleus (two polar nuclei)
C) One male gamete and one synergid cell D) Two polar nuclei and one antipodal cell

177. The transfer of pollen grains from the anther to the stigma of the exact same flower is biologically termed:

- A) Geitonogamy B) Xenogamy C) Autogamy D) Allogamy

178. The entire field of plant tissue culture relies heavily on the fundamental biological concept of 'cellular totipotency', which means:

- A) The ability of a plant cell to undergo rapid, uncontrolled mitotic division
B) The inherent capacity of a single somatic plant cell to divide and differentiate into a complete whole plant
C) The genetic resistance of a cell to bacterial and viral infections
D) The ability to mechanically fuse two different protoplasts

179. The development of 'Golden Rice', which has been genetically engineered to produce elevated levels of beta-carotene (a precursor to Vitamin A), is a prime example of:

- A) Bio-fertilization B) Bioremediation C) Biofortification D) Biomagnification

180. Which of the following organisms is widely utilized as a highly effective biofertilizer specifically in submerged paddy (rice) fields?

- A) Rhizobium B) Azotobacter C) Azolla in symbiotic association with Anabaena D) Mycorrhizal fungi

MAT

181. Select the related word pair: Book is to Reading as Fork is to:

- A) Food B) Kitchen C) Eating D) Silverware

182. In a certain code language, if 'NEPAL' is coded as 'MDOZK', how is 'KOSHI' coded?

- A) JNUGH B) JNRGH C) JNRVG D) LPTIJ

183. Pointing to a photograph, Ram said, 'I have no brother or sister, but that man's father is my father's son.' Whose photograph was it?

- A) His father's B) His own C) His son's D) His nephew's

184. A person faces North, turns 45 degrees clockwise, then 135 degrees anticlockwise, and finally 180 degrees clockwise. Which direction is he facing now?

- A) East B) West C) North D) South

185. Which of the following Venn diagrams correctly represents the relationship between: Doctors, Lawyers, Professionals?



186. What is the average of the first five prime numbers?

- A) 5.2 B) 5.4 C) 5.6 D) 5.8

187. If 20% of 'a' = 'b', then b% of 20 is the same as:

- A) 4% of a B) 5% of a C) 20% of a D) None of the above

188. A merchant sells two articles for Rs. 4,000 each. On one, he gains 20% and on the other, he loses 20%. What is his overall profit or loss percentage?

- A) No profit, no loss B) 4% Profit C) 4% Loss D) 2% Loss

189. How many 3-digit even numbers can be formed from the digits 1, 2, 3, 4, 5 if repetition is not allowed?

- A) 12 B) 24 C) 36 D) 60

190. Worker A can complete a task in 15 days. Worker B is 50% more efficient than Worker A. How many days will Worker B take?

- A) 8 days B) 10 days C) 12 days D) 22.5 days

191. Statements: All cats are dogs. Some dogs are birds. Conclusion: Some cats are birds.

A) The conclusion is True B) The conclusion is False C) Probably True D) Data Inadequate

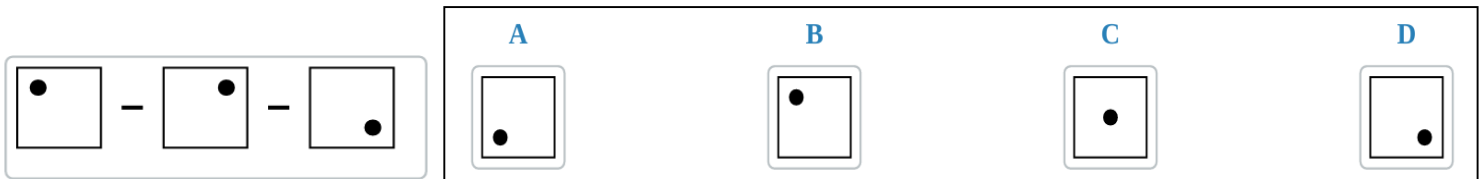
192. Statement: 'Invest in Koshi Province for better returns' - an advertisement. Which assumption is implicit?

A) Only I B) Only II C) Both I and II D) Neither I nor II

193. Event 1: Price of apples increased. Event 2: People buy fewer apples.

A) Event 1 causes Event 2 B) Event 2 causes Event 1 C) Independent causes D) Independent effects

194. Select the figure that comes next in the logical series:



195. Statement: A severe drought has been reported in rural areas. Which course of action follows?

I. The government should immediately relocate all residents to the city.

II. The government should ensure the supply of drinking water and food.

A) Only I B) Only II C) Both I and II D) Neither I nor II

196. Choose the correct mirror image of the given figure.



197. Find the odd figure out based on geometric classification.

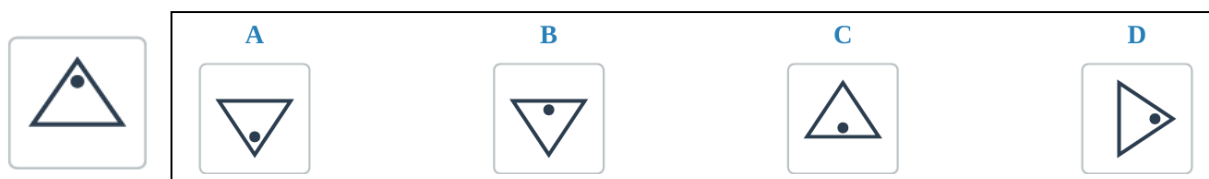


198. Identify the missing figure (?) that completes the visual matrix

logic.



199. Choose the correct water image of the given figure.



200. A square transparent sheet is folded along the dotted line. Which option represents the folded pattern?

